

War Carbon Emissions Dashboard

Methodology v1.0

WCED Scientific Steering Committee

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1 Scope

This document specifies methodology version `v1.0` of the War Carbon Emissions Dashboard (WCED). V1 quantifies CO₂ released by the combustion of hydrocarbons at oil and fuel infrastructure (refineries, crude and product depots, petrochemical complexes) inside the Iran–Gulf theatre of the 2026 Iran–Israel–US war, from 2026-02-28 onward. It does *not* cover destroyed-building embodied carbon, munitions embodied carbon, combat aviation fuel, shipping rerouting, reconstruction, or non-carbon ecological damage. Each of those is deferred to a later version (see `docs/V1_PLAN.md`, Section “How V1 Evolves”). Numbers in V1 are estimates of CO₂ released from observed fires; they are not attributed to specific belligerents.

2 Data Model

Every quantitative output crosses module boundaries as one of three Pydantic v2 types:

Distribution Carries `p5`, `p50`, `p95`, the underlying Monte Carlo `samples`, and a `provenance_id`. Emission estimates are *never* returned as scalars. Scalar arithmetic on a `Distribution` uses `apply_scalar(factor, provenance_id=factor_record_id)` so that the scalar’s own source is recorded; silent inheritance of the parent provenance is forbidden.

ProvenanceRecord Records `produced_by`, `inputs` (list of upstream provenance IDs), `method` (versioned string), `parameters` (free-form), `produced_at`, and an optional `confidence_label`. Every number in the database is reachable from a chain of these records terminating in a primary source.

ConfidenceLabel One of `CONFIRMED`, `VERIFIED`, `REPORTED`, `SUSPECTED`, `CLAIMED`. Assigned by the verification pipeline (§4) and propagated forward into every downstream estimate.

3 Emission Calculations

V1 produces two independent emission estimates per fire event — a top-down FRP-based estimate (§3.3) and a bottom-up inventory-based estimate (§3.4) — and reconciles them (§3.5). All scalar coefficients are loaded from `data/emission_factors.yaml`; all event/facility-level priors are loaded from `data/parameter_distributions.yaml`.

§3.1–3.2 (reserved)

Sections 3.1 (facility registry) and 3.2 (factor tables) are documented in the YAML files themselves and not duplicated here.

3.1 FRP-based method

Fire Radiative Power (FRP) observations from VIIRS, MODIS, and Sentinel-3 SLSTR are clustered into an event-level time series. The *raw* Fire Radiative Energy is the trapezoidal integral of the observed overpass samples:

$$I_{\text{raw}} = \int_{t_0}^{t_1} \text{FRP}(t) dt. \quad (1)$$

The *corrected* Fire Radiative Energy I_{FRE} used by all downstream equations applies two multiplicative corrections:

$$I_{\text{FRE}} = k_{\text{ext}} \cdot d \cdot I_{\text{raw}}, \quad (2)$$

where d is the burn duty cycle (`burn_duty_cycle`, $\text{triangular}(0.4, 0.7, 0.95)$) accounting for fire activity in unobserved intervals between overpasses, and k_{ext} is the FRP extrapolation factor (`frp_extrapolation_factor`, $\mathcal{N}(1.0, 0.15)$) correcting residual bias when the overpass sampling is sparse. Both d and k_{ext} are sampled independently per event in the Monte Carlo (§5); they must *never* be pre-baked into I_{raw} , since collapsing them to point values would understate the p5–p95 width of every FRP-based estimate. I_{raw} and I_{FRE} both have units of MJ.

The fuel mass combusted and resulting CO_2 mass are then

$$m_{\text{fuel}} = \alpha \cdot I_{\text{FRE}}, \quad (3)$$

$$m_{\text{CO}_2} = m_{\text{fuel}} \cdot \beta \cdot r, \quad (4)$$

with α the FRP-to-combustion-rate coefficient (`frp_to_combustion_rate`, $\mathcal{N}(0.368, 0.05)$ kg/MJ; Wooster et al. 2005), $\beta = 44/12 \cdot f_C$ the stoichiometric $\text{C} \rightarrow \text{CO}_2$ conversion with $f_C \approx 0.86$ the carbon mass fraction of crude (AP-42), and r the carbon recovery as CO_2 (`carbon_recovery_as_co2`, $\text{triangular}(0.92, 0.96, 0.98)$; Hobbs & Radke 1992). The product $\beta \cdot r$ collapses to the familiar ≈ 3.15 kg CO_2 per kg crude burned used by the CCI Iran brief when central values are substituted.

3.2 Inventory-based method

For facilities with a curated capacity C (barrels) in the `data/facilities/` registry, the bottom-up estimate is

$$m_{\text{CO}_2} = C \cdot \phi \cdot \psi \cdot \text{EF}, \quad (5)$$

where ϕ is the fraction of nameplate capacity present at strike (`facility_inventory_at_strike`, $\text{uniform}(0.3, 0.9)$), ψ is the destroyed fraction observed from before/after Sentinel-2 imagery (event-specific; default $\text{triangular}(\hat{\psi} - 0.15, \hat{\psi}, \hat{\psi} + 0.15)$ truncated to $[0, 1]$), and EF is the appropriate combustion emission factor:

- Crude depots / refinery crude units: `crude_oil_combustion` = 0.425 t CO_2 /barrel (EPA AP-42 §1.3; CCI Iran brief 2026).
- Refined-product depots: `refined_product_combustion` = 0.430 t CO_2 /barrel (IPCC 2006 GL Vol. 2 Ch. 1 Table 1.3).

Each factor entry in `emission_factors.yaml` carries an `applicable_facility_types` list. `quantify/inventory.py` must validate that the selected factor is applicable to the event’s facility type and raise if not; applying a crude factor to a gas-processing facility silently produces wrong numbers.

3.3 Reconciliation

Both estimates are Monte Carlo distributions. The reconciliation rule is:

1. Compute $\rho = m_{\text{CO}_2}^{\text{inv,p50}} / m_{\text{CO}_2}^{\text{frp,p50}}$.
2. If $0.5 \leq \rho \leq 2.0$: the two estimates agree. Publish both, flag the event `reconciled_ok`, and display the *envelope distribution* `Distribution.union(Dfrp, Dinv)` as the headline number (p5/p50/p95 over pooled samples).
3. If $\rho < 0.5$ or $\rho > 2.0$: flag `reconciliation_review`. Both estimates are stored; neither is promoted to the public dashboard until editorial review (§4.4) resolves the discrepancy. Publish only after a written reviewer note enters the changelog.
4. If a third, “reported” value exists (CEOBS / CCI / government brief), it is recorded as a `Claimed`-confidence cross-check and never enters the headline arithmetic.

4 Verification and Confidence Assignment

Each candidate fire event is independently assessed against three evidence streams: FIRMS persistence (number of overpasses with a hotspot at the location), Sentinel-2 optical classification (was a clear-sky scene found, and did the classifier label it `CONFIRMED_FIRE`), and ACLED corroboration (one or more armed events within the spatio-temporal window). The assignment is implemented in `wced/verify/confidence.py` and reproduced as Table 1.

Table 1: Confidence label decision table (v1.0).

Label	FIRMS persistent (≥ 2 passes)	S2 confirms fire	ACLED match	Notes
CONFIRMED	yes	yes	yes	all three streams agree
VERIFIED	yes	yes	no	satellite-only, no ACLED
REPORTED	yes	no	any	cloud/no-scene or S2-only
SUSPECTED	no	–	any	ACLED-only <i>never</i> a headline
SUSPECTED	no	–	no	single overpass, weak evidence
CLAIMED	–	–	–	news/state source only

The persistence threshold is fixed at ≥ 2 overpasses (`_PERSISTENT_MIN_OVERPASSES`). The ACLED-only case is intentionally non-promoting: an armed event near a facility without a FIRMS hotspot may be a near-miss, a non-incendiary strike, or a mis-geocoded record. Such candidates are routed to editorial review with a logged warning, never auto-confirmed.

Only `CONFIRMED` and `VERIFIED` events feed the headline dashboard total. `REPORTED` events appear on the map with a distinct marker. `SUSPECTED` and `CLAIMED` events stay in the pending-review queue.

5 Uncertainty Quantification

Every emission estimate is the output of a Monte Carlo simulation with $N = 10,000$ samples. The parameter PDFs are loaded verbatim from `emission_factors.yaml` and `parameter_distributions.yaml`; Table 2 reproduces the central values used in v1.0.

Table 2: Monte Carlo parameter priors (methodology v1.0).

Parameter	Distribution	Units	Source
<code>crude_oil_combustion</code>	triangular(0.405, 0.425, 0.445)	tCO ₂ /barrel	EPA AP-42 §1.3
<code>refined_product_combustion</code>	triangular(0.410, 0.430, 0.455)	tCO ₂ /barrel	IPCC 2006 GL Vol. 2
<code>frp_to_combustion_rate</code>	$\mathcal{N}(0.368, 0.05)$	kg/MJ	Wooster et al. 2005
<code>carbon_recovery_as_co2</code>	triangular(0.92, 0.96, 0.98)	–	Hobbs & Radke 1992
<code>burn_duty_cycle</code>	triangular(0.40, 0.70, 0.95)	–	expert prior (Kuwait ’91; Al
<code>facility_inventory_at_strike</code>	uniform(0.30, 0.90)	–	expert OSINT prior
<code>frp_extrapolation_factor</code>	$\mathcal{N}(1.0, 0.15)$	–	internal MODIS/VIIRS calib

Sampling rules:

- Triangular distributions are sampled on [low, high] with mode at `value`; `low/high` default to `uncertainty_low/uncertainty_high` when explicit fields are absent.
- Normal distributions use the listed `sigma` directly; the documented `uncertainty_low/high` are recorded for reviewer cross-check only and are *not* used as truncation bounds.
- Uniform distributions use strict `low/high` support.
- Parameter samples are drawn independently per event (no shared random state across events; correlation across facilities is a V2 question).
- Aggregation across events: per-event `samples` arrays are summed element-wise to yield an aggregate sample array, then p5/p50/p95 are computed on the sum.

Cache keys for any persisted Monte Carlo output must include the `methodology_version` string. Recomputing all estimates after a version bump is a deliberate operation, never automatic.

6 Worked Example: Shahrān Depot

Inputs.

- Facility: Shahrān fuel depot (Tehran), nameplate capacity $C = 5.00 \times 10^5$ barrels (crude/refined-product blend; treat as crude for V1).
- Observed destroyed fraction $\hat{\psi} = 0.40$ (Sentinel-2 before/after; triangular(0.25, 0.40, 0.55)).
- Raw FRP integral $I_{\text{raw}} = 8.5 \times 10^7$ MJ (≈ 85 TJ) from trapezoidal interpolation of 4 VIIRS/MODIS overpasses spanning an 18-hour window (mean FRP ≈ 1.3 GW, consistent with a major depot fire).
- Confidence label **CONFIRMED** (FIRMS $n_{\text{passes}} = 4$; S2 **CONFIRMED_FIRE**; 1 ACLED match within ± 24 h).

FRP-based estimate (point values). Apply the corrections of Eq. (2) at prior central values ($k_{\text{ext}} = 1.0$, $d = 0.70$), then Eqs. (3)–(4):

$$\begin{aligned} I_{\text{FRE}} &= 1.0 \cdot 0.70 \cdot 8.5 \times 10^7 = 5.95 \times 10^7 \text{ MJ}, \\ m_{\text{fuel}} &= 0.368 \cdot 5.95 \times 10^7 = 2.19 \times 10^7 \text{ kg} \approx 2.19 \times 10^4 \text{ t fuel}, \\ m_{\text{CO}_2}^{\text{frp}} &= 2.19 \times 10^7 \cdot 3.15 \approx 6.90 \times 10^7 \text{ kg} = \mathbf{6.90 \times 10^4 \text{ tCO}_2}. \end{aligned}$$

Inventory-based estimate (point values). From Eq. (5) with $\phi = 0.60$ (uniform midpoint) and $\psi = 0.40$:

$$m_{\text{CO}_2}^{\text{inv}} = 5.00 \times 10^5 \cdot 0.60 \cdot 0.40 \cdot 0.425 = 5.10 \times 10^4 \text{ tCO}_2.$$

Reconciliation. $\rho = 5.10 \times 10^4 / 6.90 \times 10^4 \approx 0.74$, inside the $[0.5, 2.0]$ band, so the event is flagged `reconciled_ok` and the envelope distribution is published. *Reviewer note:* events where ρ falls within 10 % of either boundary (i.e. $\rho \in [0.50, 0.55] \cup [1.82, 2.00]$) should receive extra editorial scrutiny — a small shift in the destroyed-fraction prior or the burn duty cycle could push them into `reconciliation_review`.

Monte Carlo result ($N = 10,000$, **methodology v1.0; reproducible fixture values used as test targets**).

Estimate	p_5 (tCO ₂)	p_{50} (tCO ₂)	p_{95} (tCO ₂)
FRP-based	3.5×10^4	6.9×10^4	1.15×10^5
Inventory-based	2.4×10^4	5.1×10^4	9.2×10^4
Envelope (headline)	2.6×10^4	6.0×10^4	1.10×10^5

These values are the fixtures expected by `tests/quantify/test_shahran_worked_example.py`; any change in sampler, RNG seed, or parameter priors must update both the methodology version string and this table.

References

- [1] U.S. Environmental Protection Agency. *AP-42: Compilation of Air Emissions Factors*, Section 1.3 (fuel oil combustion).
- [2] IPCC. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*, Volume 2, Chapter 1, Table 1.3.
- [3] Wooster, M.J., Roberts, G., Perry, G.L.W., Kaufman, Y.J. (2005). Retrieval of biomass combustion rates and totals from fire radiative power observations. *Journal of Geophysical Research*, 110, D24311.
- [4] Hobbs, P.V., Radke, L.F. (1992). Airborne studies of the smoke from the Kuwait oil fires. *Science*, 256(5059), 987–991.
- [5] Carbon Conflict Initiative. *Iran 14-day brief* (2026).
- [6] Conflict and Environment Observatory (CEOBS), incident records 2026.